

TABLE I
Brands Tested and Ranking

Rank	Brand	Model	Warranty/guarantee (in years)
1	Inalsa	Easy Cook LX	1
2	Kenwood	IH350	1
3	Bajaj	Platini PX130K	1
3	Morphy Richards	Chef xpress 100	2
4	Westinghouse	WKICLBC20	3
4	Philips	HD4907	1
5	Havells	Insta Cooks	2
5	Prestige	PIC3.OV2	1
6	Usha	Cookjoy	1
7	Glen	GL3070	1
8	Sunflame	SF1C01	1
9	Khaitan	KIC4095	1

where the energy loss is more, induction heating heats only the vessel placed over it. So the air around the vessel does not become hot, keeping the kitchen cool. Moreover, an inbuilt fan blows cool air through the electrical systems within the cooker and makes it only a little warmer than ambient temperature. Also, as the surface of the cooktop gets heated only if it is in contact with the vessel, the possibility of burn injury is significantly less.

Performance tests

Boiling time. To find out the exact warm-up time (min:sec), one litre of potable water with an initial temperature of 30°C was raised to 100°C in a recommended pan at boil mode (at maximum power), and the time taken by each cooker was recorded. The testing team found that the warm-up time was lowest for high-wattage Inalsa cooker and highest for low-wattage Glen cooker. This confirms that the heating time is inversely proportional to wattage of the product.

After water, the testing team measured the time taken to bring milk to boil. A litre of milk initially taken at room temperature (around 27°C) was raised to 100°C in a recommended pan at boil mode and the time taken

by each cooker was recorded. In this test too, Inalsa took the lowest time (2.31 minutes) and Glen (3.59 minutes), the maximum time to bring milk to boil.

Deep frying. To test the frying capability of induction cookers, a deep-frying test was conducted. All the brands cleared this test. The *pakodas* from each cooker were evenly fried and substantially crisp.

Chapati/roti baking. Arguably, the most common usage of the gas burner in a house is to make traditional Indian *rotis*. To find out whether induction cookers have the prowess to replace gas burners, this test was conducted.

All brands passed this test—the *chapatis* baked were perfect.

Tomato soup preparation. The soup was assessed to find out whether a uniform mixture was obtained after a fixed duration of boiling.

All brands performed satisfactorily during the test—each cooker produced a proper mixture of soup.

Chila making (substituted for omelette). The next test was to test the cooker's capacity to cook a *chila* of gram flour. The test was substituted for

omelette-making test since the procedure is similar for both.

All the brands passed this test as well.

Temperature distribution. A fixed quantity of potable water was boiled in a recommended pan and the temperature measured at 17 points. The observations were assessed for any significant deviation of temperature between the points.

All brands passed this test as no significant difference was observed.

Power input. Power input was measured using a fixed quantity of water as load. Appliances were run at maximum power input mode with water placed for heating in the recommended pan.

Bajaj, Morphy Richards and Inalsa cookers exceeded their rated power consumption by 28 watts, 30 watts and 126 watts, respectively.

On the other hand, the remaining nine cookers were found to have overrated their appliances. The tests found that actual input wattage of all these cookers was lower than what they claimed. The difference found was between 60 watts and 399 watts. Consumers are most likely to be misled by the declared/rated wattages of induction cookers. Since most of

the models were rated for around 2000 watts, it appears to be the ideal rating that consumers can bear in mind while buying this appliance.

The maximum variation in terms of rating/claim and the result was found in Khaitan, Westinghouse and Sunflame cookers, while the lowest variation was in Bajaj and Morphy Richards cookers.

Power consumption. There are certain key factors that reflect heating performance as well as economy of operation—namely, effective surface plate area, input power, internal design of induction

TABLE II
Time Taken by Each Cooker to Boil Water and Milk

Brand	Actual input power (watts)	For water (in minutes)	For milk (in minutes)
Bajaj	1928	3:49	3:10
Glen	1630	4:44	3:59
Havells	1790	4:05	3:29
Inalsa	2126	2:58	2:31
Westinghouse	1710	3:32	2:59
Kenwood	1840	3:53	3:18
Khaitan	1501	4:41	3:58
Morphy Richards	1830	3:59	3:23
Philips	1921	3:48	3:41
Prestige	1831	3:33	3:02
Usha	1725	4:27	3:45
Sunflame	1612	4:08	3:34